

Varen	6869	AgSp1.1	serine	rm2	1	PGTTPGTVTGRDGKPK.KFIVPNGAFTTPGSFPGP.DGEPIS.VEPAGPGTPPGIETDSDGNKKK.....LVVPTTKGPGG.Y	75
Varen	6869	AgSp1.2	serine	rm2	1	PGTTPGVVVTGPDGQPVVKYIVPREAFSTPGSIPGP.HGKRVH.VKPAGPGTPGAKTSDGDVKS.....IVLNTTPKESG..P	74
Varen	6869	AgSp1.1	serine	rm2	2	PGTTPGTVTGRDGKPKK.....FIVP.....	21
Varen	6869	AgSp1.2	serine	rm2	2	PGTTPGVVVTGPDGQPVKYIVPREAFSTPGSIPGP.HGKRVH.VKPAGPGTPGAKTSDSDGVKS.....IVLNTTPKESG..P	75
Varen	6869	AgSp1.2	serine	rm2	3	PGTTWGVVTGPDGQ.....	14
Varen	6869	AgSp1.1	serine	rm3	1	PGTTPGIVTGPDGQPV.QFIVPRGAFRTPGSIPGP.HGKRVH.VRPFPGPRTPGAKTSDSDGDISS.....IDLPTSTPVNSY..P	74
Varen	6869	AgSp1.2	serine	rm3	1	PGTTPGFVVTNSDGQPT.KFIVPRGAFRTPGTIIPGP.HGKRVH.VRPAGPGTPGAKTSDSDGDISS.....IVLPTTNKGSE..P	74
Varen	6869	AgSp1.2	serine	rm3	2	PGTTPGVVVTNSDGQPT.KFIVPRGAFRTPGTIIPGP.HGKRVH.VRPAGPGTPGAKTSDSDGDISS.....IVLPTSTPKSSD..P	74
Varen	6869	AgSp1.1	serine	rm3	2	PGTTPGVVVTGPDGQPV.QFITPRGAFRTPGSIPGP.HGKRVH.VRPAGPGTPGAKTSDSDGVDS.....IVFSPTPADSY..P	74
Varen	6869	AgSp1.1	serine	rm3	3		0
Varen	6869	AgSp1.2	serine	rm3	3	PGTTPGVVVTNSDGQPV.EFIVPRGAFRTPGTIIPGP.HGKLIH.IRPAGPGTPGAKTSDSDGVDS.....IVLPTSTPKSSD..P	74
Varen	6869	AgSp1.2	serine	rm3	4	PGTTPGVVVTNSDGQPV.QFIVPRGAFRTPGTIIPGP.HGKLIH.IRPAGPGTPGAKTSDSDGVDS.....IVLPTSTPKSSD..P	74
Varen	6869	AgSp1.2	serine	rm3	5	PGTTPGVVVTNSDGQPV.EFIVPRGAFRTPGTIIPGP.HGKLIH.IRPAGPGTPGAKTSDSDGVDS.....IVLPTSTPKSSD..P	74
Varen	6869	AgSp1.2	serine	rm3	6	PGTTPGIVTNSADGQPV.QFIVPRGAFRTPGTIIPGP.HGKLIH.IRPAGPGTPGAKTSDSDGVDS.....IVLPTSTPKSSD..P	74
Varen	6869	AgSp1.2	serine	rm3	7	PGTTPGVVVTNSDGQPV.EFIVPRGAFRTPGTIIPGP.HGKLIH.IRPAGPGTPGAKTSDSDGVDS.....IVLPTSTPKSSD..P	74
Varen	6869	AgSp1.2	serine	rm3	8	PGTTPGIVTRPDGQPV.QFIVPRGAFRTPGTIIPGP.HGKLIH.IRPAGPGTPGAKTSDSDGVDS.....IVLPTSTPKSSD..P	74
Varen	6869	AgSp1.1	serine	rm3	9	RVHVRPAGPGTPGAKTSDSDGDISS.....IVLSPMPMGSG..GP	37
Varen	6869	AgSp1.2	serine	rm3	9	PGTTPGIVTGS DGQPV.KFIVPQGAFTTPGFIMGP.HGKQIH.IGPAGPGTPGAKTSDSDGVDS.....IVLPTSTPKSSD..P	74
Varen	6869	AgSp1.1	serine	rm3	10	PGTTPGVVVTGPDGQPV.KFIVPQGAFTTPGYIPGP.HGKRVH.VGPAGPGTPGAKTSDSDGDISS.....IVLSPMPMGSG..P	74
Mg TRINITY	DN1546	c0	g1	serine	rm 1	TPGAKTSDSDGVDS.....IVLPTSTPKSSD..P	26
Mg TRINITY	DN1546	c0	g1	serine	rm 2	PGTTPGVVVTGPDGQPV.KYIVPQGAFTTPGSIIAGP.HGKRVH.VKPAGPGTPGAKTSDSDGVDS.....IVLPTSTPKSSD..P	74
Mg TRINITY	DN1546	c0	g1	serine	rm 3	PGTTPGVVVTGPDGQPV.KYIVPQGAFTTPGSIIAGP.HGKRVH.VKPAGPGTPGAKTSDSDGVDS.....IVLPTSTPKSSD..P	74
Mg TRINITY	DN1546	c0	g1	serine	rm 4	PGTTPGVVVTGPDGQPV.KYIVPQGAFTTPGSIIAGP.HGKRVH.VKPAGPGTPGAKTSDSDGVDS.....IVLPTSTPKSSD..P	74
Mraby	7048	AgSp1	serine	rm1a	1	PGTTPGAETGPDGNVDK.....IVLPTSTVRPEPQP	31
Mraby	7048	AgSp1	serine	rm1b	1	PGTTPGTVTGPDGKPS.EFIVPNGAFTTPGSIPGS.DGKPIH.VEPAGPGTPGAETGPDGKVNK.....IVLPTSTPKAP..	72
Mraby	7048	AgSp1	serine	rm1a	3	PGTTPGVVVTGPDGQPV.QFIVPQGAFTTPGSIPGP.NSKPVH.VRPAGPGTPGAKTSDSDGSIIQS.....IVLPTSTPKPSG..P	74
Mraby	7048	AgSp1	serine	rm1a	12	FIVPQGAFTTPGSIPGP.KGKPVH.VRPAGPGTPGAKTSDSDGNIDS.....IVLPTSTPKPSR..P	57
Mraby	7048	AgSp1	serine	rm1a	14	FIVPQGAFTTPGSIPGP.DGKQVH.VRPAGPGTPGAK.....	36
Mraby	7048	AgSp1	serine	rm1a	25		0
Mraby	7048	AgSp1	serine	rm1a	26	PGTTPGVVVTGPGGKPL.R.....	17
Mraby	7048	AgSp1	serine	rm1a	27	CPGGQ.....PRPQSPG.....	12
Mraby	7048	AgSp1	serine	rm1a	28	PGTTPGTVTGP.DGNPV.KFIVPQGAFTTPGSISGP.NGKPIH.VRPALPGTTPGAK.....	53
Mraby	7048	AgSp1	serine	rm1a	37		0
Mraby	7048	AgSp1	serine	rm1a	38	PGTTPGVVVTGPGGQPL.RFIVPQGAFTTPGSIPGP.NGKQIH.VRPAGPGTPGAKTSDSDGNIDS.....IVLPTSTPKSSG..P	74
Ncruc	58532and13532	AgSp1	serine	rm	87	...SPETSNGK.....KCTGENPQNANNS.LGP.LQPDSTTTTSGSQPCNPMGPSCPGGQ.....TTPGSQGPV.FDF	61
Ncruc	58532and13532	AgSp1	serine	rm	88	PGTTPGFVVTGPDGQPV.QFIVPQGAFTTPGTIKGP.HGKPIH.VKPAGPGTPGAKTSDSDGSID.....SIVLPTTP.K....	70
Ncruc	58532and13532	AgSp1	serine	rm	89	PGTTPGVVVTGPDGQPV.QFIVPQGAFTTPGTIKGP.HGKPIH.VKPAGPGTPGAKTSDSDGSID.....SIVLPTTP.K....	70
Ncruc	58532and13532	AgSp1	serine	rm	90	PGTTPGVVVTGPDGQPV.QFIVPQGAFTTPGTIKGP.HGKPIH.VRPAGPGTPGAKTSDSDGSID.....SIVLPTTP.KGSG..P	74
Ncruc	58532and13532	AgSp1	serine	rm	91	PGTTPGVVVTGPDGQPV.QFIVPQGAFTTPGTIKGP.HGKPIH.VGPAGPGTPGAKTSDSDGSID.....SIVLPTTP.KGSG..	73
Ncruc	58532and13532	AgSp1	serine	rm	92	PGTTPGIVTGPDGQPV.KFIVPQGAFTTPGTIPGP.HGKPIH.IGPAGPGTPGAKTSDSDGSID.....SIVLPTTP.KGSG..	73
Ncruc	58532and13532	AgSp1	serine	rm	93	PGTTPGIVTGPDGQPV.KFFVPPNGAFTTPGSIPGP.HGKPIH.IGPAGPGTPGAKTSDSDGDIS.....EIDLPTSTP.SPAP..P	74
Msag	19329	AgSp1.1	serine	rm	1	AFTTPGSIIAGP.HGKRVH.VKPAGPELHQE.LKPSDGDVSS.....IVLPTSTPKSSG..G.P	50
Msag	19329	AgSp1.1	serine	rm	2	PGTTPGVVVTGPDGQPV.VKYVVPQ.EPYDSWIYCRTS.BKRVH.VKPAGPGTPGAKTSDSDGVSS.....IVLPTSTPKSSG...P	72
Msag	19329	AgSp1.1	serine	rm	3	PGTTPGVVVTGPDGQPV.VKYVVPQGAFTTPGSIIAGP.HGKRVH.VKPAGPGTPGAKTSDSDGVSS.....IVLPTSTPKASG...P	73
Msag	19329	AgSp1.1	serine	rm	4	PGTHLAVVTGPDGQPV.VKYVVPQVAFSTTPGSIIAGP.HGKRVHAREPAGPGTPGAKTSDSDGVSS.....IVLPTSTPKSSG...P	74
Msag	19329	AgSp1.1	serine	rm	5	PGTTPGVVVTGPDGQSQFKYVVPQGAFTTPGSIIAGP.HGKRVH.VKPAGPGTPGAKTSDPDGDVSS.....IVLPTSTPKSSG...P	74
Msag	19329	AgSp1.1	serine	rm	6	PGTTPGVVVTGPDGQPV.VKYVVPQGAFTTPGSIIAGP.HGKRVH.VKPAGPGTPGAKTSDSDGVSS.....IVLPTSTPK.....P	70
Msag	19329	AgSp1.1	serine	rm2	1	PGITPGAKTSDSDGVES.....IVVPTSTPKSSG..SP	29
Msag	19329	AgSp1.1	serine	rm2	16		0
Msag	19329	AgSp1.1	serine	rm2	17	PGITPGAKTSDSDGVES.....IVVPTSTPKSSG..DP	29
Lcorn	58650	AgSp1.2	serine	rm1	3	GSGPG.....	5
Lcorn	58650	AgSp1.2	serine	rm1	4	PGTTPGVVVTGPDGQPV.KFIVPQGAFTTPGSIIAGP.NGKPIH.VKPAGPGTPGAKTSDSDGSIDS.....IVLPTST.PK....	70
Lcorn	58650	AgSp1.2	serine	rm1	5		0
Lcorn	58650	AgSp1.2	serine	rm1	6	F.IVP	4
Lcorn	58650	AgSp1.2	serine	rm1	7		0
Lcorn	58650	AgSp1.2	serine	rm1	8		0
Lcorn	58650	AgSp1.2	serine	rm1	10	G.SGT	4
Lcorn	58650	AgSp1.2	serine	rm1	11		0
Lcorn	58650	AgSp1.2	serine	rm1	12	F.IVP	4
Lcorn	58650	AgSp1.2	serine	rm1	17		0
Lcorn	58650	AgSp1.2	serine	rm1	21		0

Varen 6869 AgSp1.1 serine rm2 1	PM.....GPGSPFGPGNMF.....GP..MR.PGRTTTPSSVEGPDGEPPIQIEPAG..	117
Varen 6869 AgSp1.2 serine rm2 1	GF.....DFTTPSHVTKGDKGPIQIVPAG..	98
Varen 6869 AgSp1.1 serine rm2 2	NG.A.....FTTPGSFPGPDGEPISVEPACPG	47
Varen 6869 AgSp1.2 serine rm2 2	GF.....DFTTPSHVTKGDKGPIQIVPAG..	99
Varen 6869 AgSp1.2 serine rm2 3	PVKGDGQPIQIVPACPG	31
Varen 6869 AgSp1.1 serine rm3 1	RY.....RFQTPSRVTKPDGEPITIVPAG..	98
Varen 6869 AgSp1.2 serine rm3 1	GV.....HFETTKAITEPDGQPIVIVPAG..	98
Varen 6869 AgSp1.2 serine rm3 2	GF.....QFQTPKQITEPDGQPIRIIPAG..	98
Varen 6869 AgSp1.1 serine rm3 2	GF.....QFQTPSRVTKPDGQPIETIVPAG..	98
Varen 6869 AgSp1.1 serine rm3 3	PG	2
Varen 6869 AgSp1.2 serine rm3 3	GY.....QFQTPKQITEPDGQPIETIPAG..	98
Varen 6869 AgSp1.2 serine rm3 4	GF.....QFQTPKHITEPDGQPIRIIPAG..	98
Varen 6869 AgSp1.2 serine rm3 5	GY.....QFQTPKKITEPDGQPIETIPAG..	98
Varen 6869 AgSp1.2 serine rm3 6	GF.....QFQTPKHITEPDGQPIRIIPAG..	98
Varen 6869 AgSp1.2 serine rm3 7	GY.....QFQTPKKITEPDGQPIETIPAG..	98
Varen 6869 AgSp1.2 serine rm3 8	GF.....QSQTPTNTGTGQPIQIIPAG..	98
Varen 6869 AgSp1.1 serine rm3 9	GF.R.....FPTPRRVPMPDGQPIQIIPAG..	61
Varen 6869 AgSp1.2 serine rm3 9	GF.....QSQTPTNTTKTEGQPIQIIPAG..	98
Varen 6869 AgSp1.1 serine rm3 10	GFQ.....	77
Mg TRINITY DN1546 c0 g1 serine rm 1	GF.Q.....SPTTPSHVTKPDGQPIQIVPAG..	51
Mg TRINITY DN1546 c0 g1 serine rm 2	GFQS.....PTTPSHVTKPDGQPIQIVPAG..	99
Mg TRINITY DN1546 c0 g1 serine rm 3	AFQS.....PTTPSHVTKPDGQPIQIVPAG..	99
Mg TRINITY DN1546 c0 g1 serine rm 4	GFQS.....PTTPSHVTKPDGQPIQIVPAG..	99
Mraby 7048 AgSp1 serine rm1a 1	Q.....TTTTKVKGPQCKPVLVVPAG..	52
Mraby 7048 AgSp1 serine rm1b 1	TGPGGQPLNPLGPGSPLRP.....GSPLGPGNLN.....ATPATTTPTVPGPGCKPLQIVPAG..	125
Mraby 7048 AgSp1 serine rm1a 3	GFN.....FQTPSPVTKPDGQPIQIVPAG..	98
Mraby 7048 AgSp1 serine rm1a 12	GK.....QKKNLPVVTQSDGQPIQIVPAG..	81
Mraby 7048 AgSp1 serine rm1a 14		36
Mraby 7048 AgSp1 serine rm1a 25	NPLPVVTQSDGQPIQIVPAG..	19
Mraby 7048 AgSp1 serine rm1a 26		17
Mraby 7048 AgSp1 serine rm1a 27	.Q.Q.....TTTPEPIPGFNCKPLQIVPAG..	35
Mraby 7048 AgSp1 serine rm1a 28		53
Mraby 7048 AgSp1 serine rm1a 37	NPLPVVTQPDGQPIQIVPAG..	19
Mraby 7048 AgSp1 serine rm1a 38	VP.....ITTT..PIPGPGPEPLQIVPAG..	96
Ncruc 58532and13532 AgSp1 serine rm 87	HTPSPITKPDGKPIQIVPAG..	81
Ncruc 58532and13532 AgSp1 serine rm 88	GSGPGF..QF.....QTPQPVTKPDGQPIQIVPAG..	98
Ncruc 58532and13532 AgSp1 serine rm 89	GSGPGF..QF.....QTPQPVTKPDGQPIQIVPAG..	98
Ncruc 58532and13532 AgSp1 serine rm 90	G.....FPTTPKG.....SGPGF..QF.....QTPQPVTKPDGQPIQIVPAG..	109
Ncruc 58532and13532 AgSp1 serine rm 91	PGF..NF.....ETPRPAPQ..SGGPIQIVPAG..	97
Ncruc 58532and13532 AgSp1 serine rm 92	PGY..NF.....PTPRPAPQ..SGGPIQIVPAG..	97
Ncruc 58532and13532 AgSp1 serine rm 93	QP.A.....NPTTPTDVRGFPQGSPIPIVPAG..	99
Msag 19329 AgSp1.1 serine rm 1	G.....F.....QSPTPSQVTKPDGQPIQIVPAG..	74
Msag 19329 AgSp1.1 serine rm 2	PG.....F.....HTPTPSQVTKPDGQPIQIVPAG..	97
Msag 19329 AgSp1.1 serine rm 3	PG.....F.....QSPTPSQVTKPDGQPIQIVPAG..	98
Msag 19329 AgSp1.1 serine rm 4	PG.....F.....RTPTLSQVTKPDGQPIQIVPAG..	99
Msag 19329 AgSp1.1 serine rm 5	PG.....F.....QSPTPSQVTKPDGQPIQIVPAG..	99
Msag 19329 AgSp1.1 serine rm 6		70
Msag 19329 AgSp1.1 serine rm2 1	GF.Q.....SPTPRRVTKPACQPIETIVAAGPG	55
Msag 19329 AgSp1.1 serine rm2 16	YVVPQGAFTTPGSFAGVHKRVRVHPAG..	28
Msag 19329 AgSp1.1 serine rm2 17	GF.R.....APTTPSRATEFTGQPIQIVTTG..	53
Lcorn 58650 AgSp1.2 serine rm1 3	.F.Q.....FQTPQPITKPGGQPIQIVPAG..	28
Lcorn 58650 AgSp1.2 serine rm1 4	G.....SRPGF..QF.....QTPQPITKSGGQPIQIVPAG..	98
Lcorn 58650 AgSp1.2 serine rm1 5	PG	2
Lcorn 58650 AgSp1.2 serine rm1 6	QG.A.....FTTSGSIPGPNCKPIHVKPACPG	30
Lcorn 58650 AgSp1.2 serine rm1 7	HGCGQPIQIVPAG..	12
Lcorn 58650 AgSp1.2 serine rm1 8	PG	2
Lcorn 58650 AgSp1.2 serine rm1 10	GF.Q.....FQTPQPITKPGCGQPIQIVPAG..	28
Lcorn 58650 AgSp1.2 serine rm1 11	PG	2
Lcorn 58650 AgSp1.2 serine rm1 12	QG.A.....FTTPGSILGPNCKPIHVKPACPG	30
Lcorn 58650 AgSp1.2 serine rm1 17	FIVPQGAFTTPGSIPGPNCKPIR.....	23
Lcorn 58650 AgSp1.2 serine rm1 21	SGGQPIQIVPAG..	12

Varen 6869 AgSp1.1 serine rm2 1		117
Varen 6869 AgSp1.2 serine rm2 1		98
Varen 6869 AgSp1.1 serine rm2 2	TTPGIETDSDGNIKKVVVPTTT.....	69
Varen 6869 AgSp1.2 serine rm2 2		99
Varen 6869 AgSp1.2 serine rm2 3	TTLGVVTGPDGQPVKPRAVTEPDGEP.....IQIIPAG	64
Varen 6869 AgSp1.1 serine rm3 1		98
Varen 6869 AgSp1.2 serine rm3 1		98
Varen 6869 AgSp1.2 serine rm3 2		98
Varen 6869 AgSp1.1 serine rm3 2		98
Varen 6869 AgSp1.1 serine rm3 3	TTPGVVTGPDGQPVQFIVPRGAFRT.....	27
Varen 6869 AgSp1.2 serine rm3 3		98
Varen 6869 AgSp1.2 serine rm3 4		98
Varen 6869 AgSp1.2 serine rm3 5		98
Varen 6869 AgSp1.2 serine rm3 6		98
Varen 6869 AgSp1.2 serine rm3 7		98
Varen 6869 AgSp1.2 serine rm3 8		98
Varen 6869 AgSp1.1 serine rm3 9		61
Varen 6869 AgSp1.2 serine rm3 9		98
Varen 6869 AgSp1.1 serine rm3 10		77
Mg TRINITY DN1546 c0 g1 serine rm 1		51
Mg TRINITY DN1546 c0 g1 serine rm 2		99
Mg TRINITY DN1546 c0 g1 serine rm 3		99
Mg TRINITY DN1546 c0 g1 serine rm 4		99
Mlaby 7048 AgSp1 serine rm1a 1		52
Mlaby 7048 AgSp1 serine rm1b 1		125
Mlaby 7048 AgSp1 serine rm1a 3		98
Mlaby 7048 AgSp1 serine rm1a 12		81
Mlaby 7048 AgSp1 serine rm1a 14		36
Mlaby 7048 AgSp1 serine rm1a 25		19
Mlaby 7048 AgSp1 serine rm1a 26		17
Mlaby 7048 AgSp1 serine rm1a 27		35
Mlaby 7048 AgSp1 serine rm1a 28		53
Mlaby 7048 AgSp1 serine rm1a 37		19
Mlaby 7048 AgSp1 serine rm1a 38		96
Ncruc 58532and13532 AgSp1 serine rm 87		81
Ncruc 58532and13532 AgSp1 serine rm 88		98
Ncruc 58532and13532 AgSp1 serine rm 89		98
Ncruc 58532and13532 AgSp1 serine rm 90		109
Ncruc 58532and13532 AgSp1 serine rm 91		97
Ncruc 58532and13532 AgSp1 serine rm 92		97
Ncruc 58532and13532 AgSp1 serine rm 93		99
Msag 19329 AgSp1.1 serine rm 1		74
Msag 19329 AgSp1.1 serine rm 2		97
Msag 19329 AgSp1.1 serine rm 3		98
Msag 19329 AgSp1.1 serine rm 4		99
Msag 19329 AgSp1.1 serine rm 5		99
Msag 19329 AgSp1.1 serine rm 6		70
Msag 19329 AgSp1.1 serine rm2 1	TTPGVVTGPDGQPVRYVVPQGAFTTPGSFAGLHGK.....	90
Msag 19329 AgSp1.1 serine rm2 16		28
Msag 19329 AgSp1.1 serine rm2 17		53
Lcorn 58650 AgSp1.2 serine rm1 3		28
Lcorn 58650 AgSp1.2 serine rm1 4		98
Lcorn 58650 AgSp1.2 serine rm1 5	TTPGVVTGADGQPVK.....	17
Lcorn 58650 AgSp1.2 serine rm1 6	TTPGAK.....	36
Lcorn 58650 AgSp1.2 serine rm1 7		12
Lcorn 58650 AgSp1.2 serine rm1 8	TTPGVVTGPDGQPVK.....	17
Lcorn 58650 AgSp1.2 serine rm1 10		28
Lcorn 58650 AgSp1.2 serine rm1 11	TTPGVVTGPDGQPVK.....	17
Lcorn 58650 AgSp1.2 serine rm1 12	TTPGAK.....	36
Lcorn 58650 AgSp1.2 serine rm1 17		23
Lcorn 58650 AgSp1.2 serine rm1 21		12

 non-conserved
 similar
 $\geq 50\%$ conserved